# Jiaxin (Jeremiah) Pan

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## Research Profile

Physics undergraduate working on AI for Science, generative sampling, interpretable materials representations, and language-model reasoning. Research interests connect statistical mechanics with reproducible scientific computing and reliable machine-learning methods.

## Education

2023–2027 **B.Sc. in Physics (with Honours) — First Class Honours expected**

The Chinese University of Hong Kong, Shenzhen · Shenzhen, China

GPA: 3.61/4.0. Four-year honours degree with a required research thesis. Admitted to Economics in 2023; completed Year 1 foundation courses before transferring to the B.Sc. Physics programme in 2024 to pursue my passion for sciences.

## Exchange and Visiting Programmes

2026 (Spring) **Visiting Student**

University of Auckland · Auckland, New Zealand

Nominated by CUHK-Shenzhen; completed advanced coursework in Physics, Computer Science, and AI, including COMPSCI 361 and COMPSCI 220.

2025 (Summer) **Summer Visiting Student**

University of California, Berkeley Extension · Berkeley, CA, USA

Completed coursework in physics and neuroscience; developed cross-cultural academic communication and independent learning skills.

## Ongoing Research

Ongoing **Data-Free Generative Sampling of the Two-Dimensional XY Model**

Undergraduate Thesis, CUHK-Shenzhen

Supervised by Prof. Kai Zhou. Provisional portfolio title, not a finalized thesis title.

- Contributed to adapting Stochastic Path Sampling (SPS) to periodic XY angle variables. Training uses the XY energy and forward/backward path probabilities without pre-generated equilibrium configurations as training data. Generated paths are evaluated with path-space independence Metropolis–Hastings (MH) correction.
- Earlier exploration used supervised Flow Matching trained on Wolff-generated configurations. Wolff sampling is an independent validation reference for the current method.
- Completed a training-budget comparison across three seeds, 5,000/10,000/15,000 parameter updates, and 32/64/128 generation steps. Python/PyTorch GPU experiments run through Slurm.
- Evaluated energy, vortex counts, magnetization, spatial correlations, acceptance, effective sample size per sampling second, and consecutive rejection streaks.
- Estimated energy sampling efficiency increased in all nine seed/step settings from 5,000 to 10,000 updates; from 10,000 to 15,000, six improved and three declined. All nine settings at both later checkpoints passed predefined physical-statistics screening criteria; this supports, but does not prove, complete equilibration.
- High relative path weights can cause prolonged MH rejection streaks. Acceptance alone does not determine sampling efficiency. Analysis and report preparation are underway.

The limited 4×4 lattice study at one temperature near the expected Berezinskii–Kosterlitz–Thouless (BKT) transition does not establish thermodynamic critical behavior, a solution to critical slowing down, universal convergence, or an advantage over Wolff. SPS is an existing method.

Ongoing **Interpretable Crystal Representations for AI-Assisted Materials Design**

HSSRLM — AI for Science — Working paper

Co-first author; manuscript in preparation for planned submission to the Annual Meeting of the Association for Computational Linguistics (ACL). Not yet submitted. Framework: compact hierarchical descriptors form an interpretable interface between natural language and porous materials, including metal–organic frameworks (MOFs) and zeolites.

- Hierarchical Structure-Sequence Representation (HSSR) organizes composition, lattice parameters, symmetry, topology, building units, and selected pore-related properties into structured text. The repository implements T5 (Text-to-Text Transfer Transformer) infrastructure for bidirectional text/descriptor translation.
- Contributed to evidence-aware Crystallographic Information File (CIF)-to-descriptor preprocessing that separates known, inferred, and missing information, preserves provenance, and routes incomplete records for review.
- Contributed to multi-candidate screening with descriptor validation, node/linker/topology connectivity checks, explicit failure reporting, and descriptor-level duplicate tracking.
- Contributed to reproducible structure-equivalence audits on three publicly sourced MOF structures, examining equivalent representations and information lost by coarse descriptors. This is a limited study, not evidence of universal representation invariance.

Framework implementation and targeted reliability audits are complete; broader dataset validation and model evaluation remain in progress. HSSR omits complete atomic coordinates and a full bonding graph, so its outputs are descriptors rather than complete atomistic CIF structures. Verification establishes compatibility with implemented rules, not physical validity or synthesizability. CrysSpec (Crystal Specification) is a provisional working name. The framework also includes a 26-record metal-node registry that addresses ambiguous aliases and preserves source material for continued chemical validation.

## Additional Research Experience

2026 (Spring) **Remote Research Assistant**

INSAIT · Bulgaria / Remote

Under Dr. Jinjin Gu.

- Analysed large language model reasoning and post-training, including Surgical Post-Training (SPOT), implicit regularization, minimal-edit data rectification, and catastrophic forgetting.
- Studied the trade-off between reasoning performance and foundational knowledge retention; prepared research syntheses and contributed to seminars.

2025 (Fall) **Guided Research on Van der Waals Integration and 2D Electronics**

CUHK-Shenzhen, Prof. Qi Qian's Lab

Participated in fabrication and characterization of $MoS_2$ transistors through van der Waals integration and dry transfer of 2D heterostructures. Studied metal–semiconductor contacts and polarity modulation; gained hands-on nanofabrication and electrical characterization experience, conducted literature review, and prepared an end-of-term report.

Earlier work **Guided Research Training in Generative AI and Diffusion Models**

CUHK-Shenzhen, Prof. Kai Zhou's Group

Earlier training in score-based generative modelling and physical dynamics.

- Explored connections between stochastic differential equations and diffusion models, including the denoising score-matching objective.
- Developed simulations of a one-dimensional double-well potential and visualized generative trajectories.
- Analysed learned score fields and local stability using Jacobian matrices; compared Langevin-sampled steady-state distributions with the Gibbs distribution.

## Submitted Manuscript

Submitted **Mathematics self confidence and STEM major choice in an undergraduate survey**

Reinvention: an International Journal of Undergraduate Research

Sole author. Questionnaire study examining associations between mathematics self-confidence and STEM major choice. Submitted; awaiting editorial assessment. Separately, a preprint is publicly posted on SSRN. Presented at URC2026. URC2026 paper and slides (Zenodo). Zenodo and SSRN host the same final paper submitted to URC2026; Zenodo also includes the presentation slides.

## Conference

31 Jan 2026 **CUHK-Shenzhen Undergraduate Research Conference (URC2026)**

Second Undergraduate Research Conference · Shenzhen, China

Invited to present my original, single-author ENG2001 research project, “Self-Confidence in Math and STEM Major Choice at CUHK-Shenzhen”, at my first research conference. The study examined the relationship between mathematical self-confidence and STEM major choice. My ENG2001 instructor, Dan, described it as “a great quantitative paper” and requested permission to use an anonymised version as a classroom example. Research paper (Zenodo DOI); presentation certificate and on-site photographs on the portfolio.

## Course Projects

2025 (Fall) **Group Research on Bose–Einstein Condensation**

CUHK-Shenzhen

Group leader. Coordinated literature study of “Observation of Bose-Einstein Condensation in a Dilute Atomic Vapor”; delegated tasks, co-authored LaTeX presentation slides, and delivered the class presentation.

2025 (Spring) **Group Research on Bell Test in Quantum Physics**

CUHK-Shenzhen

Group leader. Coordinated a study of experimental verification of Bell's theorem; delegated tasks, co-authored the final report, and delivered the presentation.

2025 (Spring) **Group Research on Computational Physics: Quantum Harmonic Oscillator**

CUHK-Shenzhen

Used finite-difference discretization and matrix eigenvalue methods for the one-dimensional time-independent Schrödinger equation. Compared numerical and analytical solutions, examined convergence with grid resolution and domain width, and synthesized team members' results into a final LaTeX report. Co-authored with Mengxiang Zhu; deposited as a technical note on Zenodo in 2026. DOI: 10.5281/zenodo.23439036.

2024 (Fall) **Group Research on Semiconductor Optical Properties**

CUHK-Shenzhen

Group leader. Coordinated a study of semiconductor optical properties; conducted literature review, authored the final report, and delivered the class presentation.

## Teaching and Work Experience

Oct–Dec 2026 **Undergraduate Student Teaching Fellow (USTF)**

CUHK-Shenzhen School of Science and Engineering

Reappointed to the Student Help Room for 21 October–18 December 2026, supporting MAT1001 (Calculus), CSC1005 (Introduction to Computer Engineering), and PHY1001 (Mechanics).

Oct–Dec 2025 **Undergraduate Student Teaching Fellow (USTF)**

CUHK-Shenzhen School of Science and Engineering

Provided tutoring in the Student Help Room, a form of USTF service, for CSC1005 (Introduction to Computer Engineering), CSC1001 (Introduction to Computer Science), and PHY1001 (Mechanics), 23 October–12 December 2025. Help Room certificate available on the portfolio.

2025–2026 **Study Abroad Student Ambassador**

CUHK-Shenzhen Office of Academic Links · Australia and New Zealand Group

Served as a liaison between students and OAL for communication, safety reminders, and emergency handling; supported students' academic and living experiences, organized group activities, and contributed to outbound-study sharing sessions. Recognized by OAL for exemplary performance.

2026 (Spring) **Study Abroad Instagram Ambassador**

University of Auckland · Auckland, New Zealand

March–June 2026. Produced multimedia content for the official Study Abroad Instagram account; recognized by the Study Abroad Team as the most prolific ambassador with the highest total likes across posts. Recommendation letter available on the portfolio.

2023–2025 **Private Tutor**

Self-employed

Tutored CET-4, college entrance English, and international middle-school mathematics using the Big Ideas Math curriculum; tailored lesson plans to students' needs.

2023 (Summer) **Teaching Assistant**

New Oriental Education and Technology Group · Chongqing, China

Supported English instruction, classroom management, teaching-material preparation, and communication with parents.

## Honours and Awards

2023–2026 **Full-Tuition Scholarship**

CUHK-Shenzhen

Philip Yu Hong Wong Anita Fung Yee Leung Full-tuition Scholarship. Awarded based on outstanding performance in the 2023 Gaokao.

2026 **Dean's List**

CUHK-Shenzhen School of Science and Engineering

Recognized for academic excellence in AY2025–2026. Certificate pending issuance.

2025 **Dean's List**

CUHK-Shenzhen School of Science and Engineering

Recognized for academic excellence in AY2024–2025.

2024 **Dean's List**

CUHK-Shenzhen School of Management and Economics

Recognized for academic excellence in AY2023–2024.

2024–2025 **Shaw Overcoming Adversity Award**

CUHK-Shenzhen Shaw College

Recognized for resilience and perseverance in overcoming personal and academic challenges.

## Certification

2025 HCIA-AI: Huawei Certified ICT Associate — Artificial Intelligence. Issued by Huawei Technologies Co., Ltd.; valid through December 2028. Certificate No. 01010200181006237292 7192.

## Skills

Scientific computing: Python, PyTorch, NumPy, SciPy, Matplotlib, CUDA, Git, LaTeX; GPU experiment workflows through Slurm. Coursework includes CSC1001 (A grade), CSC1006 (machine-learning concepts), and PHY2650 (team project awarded full marks plus an excellence bonus).

Machine learning: Diffusion models, Flow Matching, and Diffusion Transformers (DiT) for AI-for-Science applications.

Collaboration: Microsoft Word, Excel, PowerPoint, Teams, and Planner.

Content creation: Instagram, WeChat Official Accounts, and TikTok; photo/video editing with iMovie and TikTok Studio.

Piano: Passed the Chinese Musicians Association amateur piano Grade 8 examination.

Chess: Recreational player; approximately 1200 Elo on Duolingo, playing against Oscar.

## Languages

English: Full professional proficiency. TOEFL iBT: 100 (Reading 29, Listening 26, Speaking 23, Writing 22), October 2025.

Mandarin: Native proficiency.

Choral pronunciation: Trained in German, Cantonese, Latin, Maori, and Japanese pronunciation through choir rehearsals and performances, with experience focused on singing.

## Extracurricular Activities

Music and the arts — An unwavering love of the arts alongside my studies and research, expressed through choral singing and public performance.

2026 (Fall) — **CUHK-SZ Chorus**

Tenor

Joined the university chorus in Fall 2026, continuing my commitment to choral music.

2026 (Spring) — **Auckland University Student Choir (AUSC)**

Tenor · University of Auckland exchange

Performed in two public concerts: the ANZAC Eve Commemorative Concert (24 April 2026) and *Oh, the Places We'll Go* (29 May 2026). Performance record; six-video concert playlist.

2025 (Fall) — **School of Music Choir, CUHK-SZ**

Tenor; Choir Class Member

Participated in the choir class and performed in the end-of-term concert, *Winter Carols*, on 12 December 2025.

2024–Present — **CUHKSZ Economics Club**

Core Management; Head of Secretary Department

- Recruited new members, led the Secretary Department team, delegated tasks, and coordinated work across club departments.
- Managed the club mailbox, relayed important information, and drafted and sent notices and event emails.
- Created registration forms with Wenjuanxing and consolidated responses for club activities.
- Contributed to EC Academic Weekly, event copy, and social media content; maintained club documents and supported event logistics.

2024–Present — **CUHKSZ Physical Society**

Management

Organized academic events and coordinated internal operations.

2023–2024 — **CUHKSZ Shaw College**

Student Helper

Created WeChat content, provided multimedia support, and assisted in campus-event organization.

2023 — **CUHKSZ School of Humanities and Social Sciences**

Research Assistant

Supported research documentation through field photography and video editing; assisted professors with data organization.